

Exercises on simple loops

1. Power

Write a program that calculates b raised to the n , where b and n are a `double` and an `unsigned int`, respectively, entered by the user.

2. First n cubes

Write a program that calculates all the "cubes" less than or equal to n , where n is an `unsigned int` entered by the user.

For example, if n is 100 then the output is:

```
1  8  27  64
```

3. Fibonacci

Write a program that calculates the first n values of Fibonacci's sequence, where n is an `unsigned int` entered by the user.

The first two values of the sequence are 1 1. The other values are obtained as sum of the two preceding values, that is, if n is greater than two:

```
fib(n) = fib(n-1) + fib(n-2)
```

4. Largest monotone subsequence

Write a program that reads an arbitrary sequence of integers and calculates the length of the largest subsequence of increasing values.

For example, if the input sequence is

```
-1 -2  3  4  7  0  1  2
```

then the output is: 4

Nested loops

1. Primes

Write a program that calculates the first n prime numbers, where n is an `unsigned int` entered by the user.

For example, if n is 6 then the output is:

```
2 3 5 7 11 13
```

2. Triangle

Write a program that draws a "triangle" of size n , where n is an `unsigned int` entered by the user.

For example, if n is 3 then the output is:

```
  *
 * * *
* * * *
```

2. Powers' grid

Write a program that calculates the grid of powers x to the y such that x and y are values ranging in $[1, n]$, where n is an `unsigned int` entered by the user.

For example, if n is 4 then the output is:

```
1 1 1 1
2 4 8 16
3 9 27 81
4 16 64 256
```